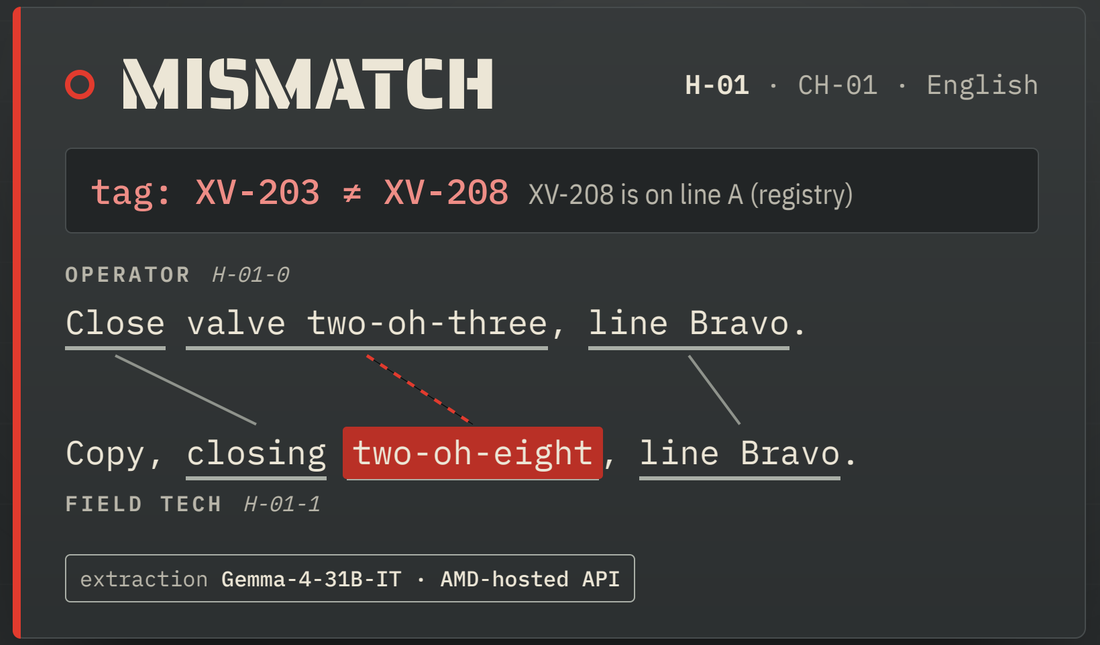
SAYAGAIN

NOBODY ON THE CHANNEL NOTICED. THE RADIO LOG DID.

Checks each read-back of a radioed safety order against the plant registry; wrong ones, and any it cannot check, reach the supervisor.




```
zsh · build/
$ jq -r '…' bench/results/text/llm-judge-r1-reaudit.json
llm_judge false-VERIFIED 2 false alarms 1/48
$ jq -r '…' bench/results/text/ladder-r1-reaudit.json
Gemma-4-31B-IT false-VERIFIED 0 false alarms 2/48
regex false-VERIFIED 0 false alarms 0/48
# 3 more rungs, false-VERIFIED 0 on each
```

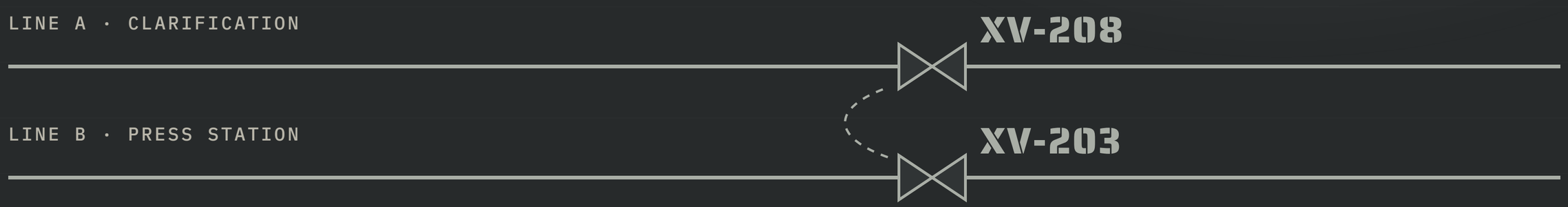
CH-01 · HOW THE CHANNEL HEARS IT TODAY

TWO LINES. NOBODY FLAGS THEM.

OPERATOR — Close valve two-oh-three, line Bravo.

FIELD TECH — Copy, closing two-oh-eight, line Bravo.

no flag · no card · the crew walks to XV-208

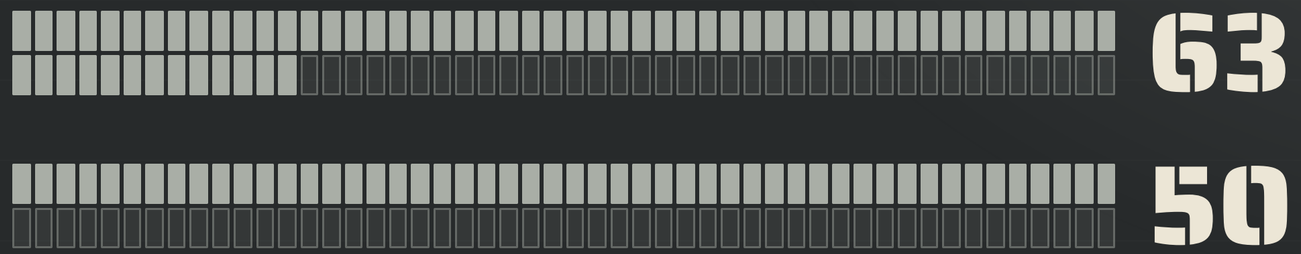


In the registry the demo runs on, XV-203 is on line B at the press station; XV-208 is on line A, in clarification.

THREE-WAY COMMUNICATION

READ-BACK IS THE CHECK. NOTHING CHECKS IT.

WRONG READ-BACKS CAUGHT, OF EVERY 100 · AVIATION, HELMKE 2021



filled: caught by the controller · empty: went through

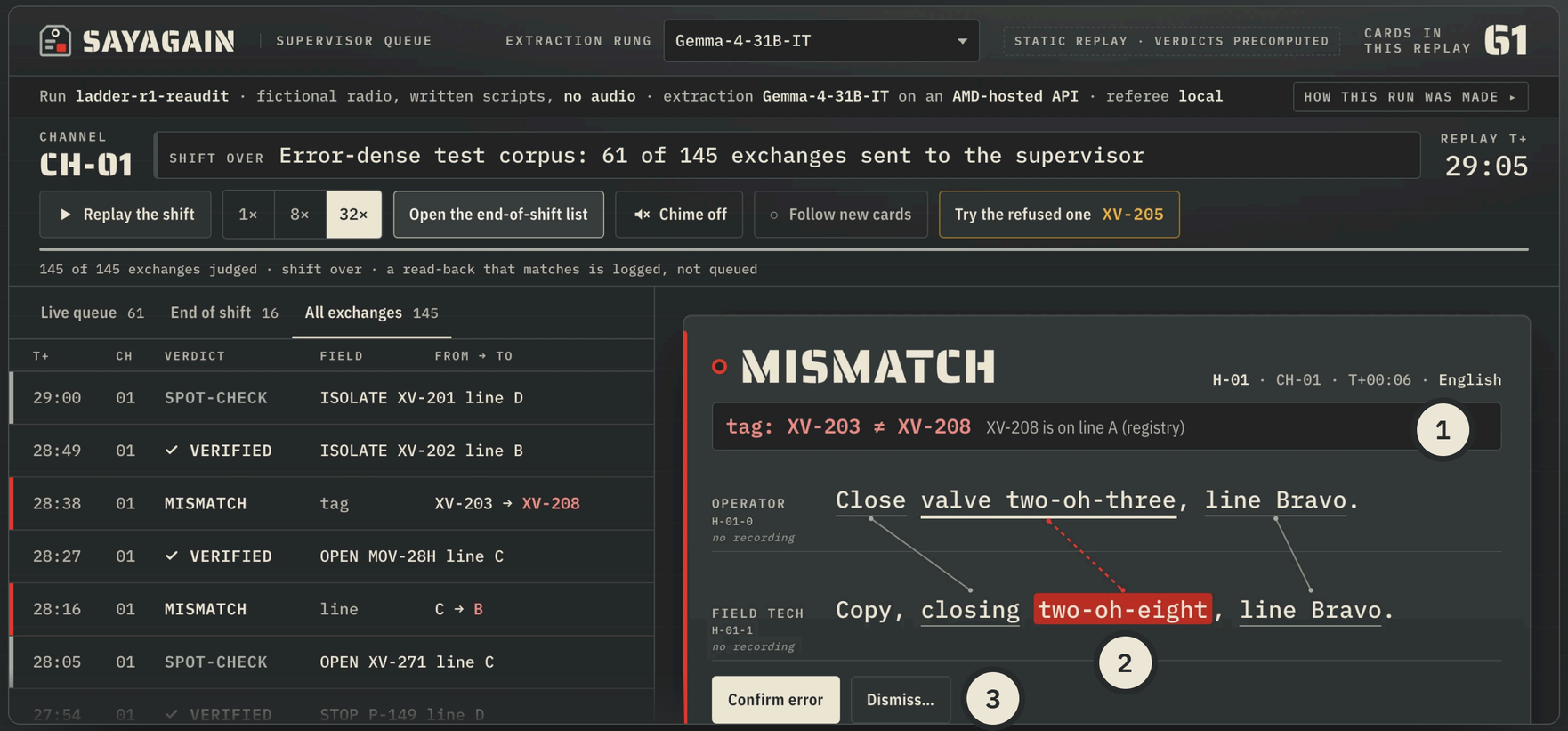
NO RATE

published for plant radio

WHY IT MATTERS · US CSB

Goodyear, Houston, 2008: a relief-path valve left closed after maintenance, the other path then blocked. **1 killed, 6 injured.**

THE SUPERVISOR GETS THE ONE WRONG WORD.



1. the diff, checked against the registry
2. the read-back word that differs
3. a human confirms or dismisses

ONE MODEL JOB · EVERYTHING ELSE IS CODE

THE MODEL POINTS. CODE DECIDES.

1 · THE EXCHANGE THE MODEL READ

ORDER

Close valve two-oh-three, line Bravo.

READ-BACK

Copy, closing two-oh-eight, line Bravo.

H-01 · written words, text·scripted

→

2 · WHAT GEMMA-4-31B-IT RETURNED

```
{
  "commands": [
    {
      "action_span": "closing",
      "line_span": "line Bravo",
      "state_span": null,
      "tag_span": "two-oh-eight"
    }
  ],
  "kind": "readback",
  "retracted_spans": []
}
```

AMD Token Factory developer.amd.com.cn

median latency 4.304 s, first pass

cost USD 4.8937e-06

→

3 · WHAT CODE DECIDED

1. instruction: grounded + covered (1 command(s))
2. read-back: grounded + covered (1 command(s))
3. audio: no gate (no token probabilities, no unprompted pass)
4. instruction: XV-203 line B (registry ok)
5. compare: tag: XV-203 ≠ XV-208

○ MISMATCH

tag: XV-203 ≠ XV-208

speech-to-text: none (written words) · **span extraction: AMD Token Factory** · referee: local, plain Python · queue: local

receipt, as stored · request_id c9647a04cfcc4eeaadbfa965a4d74c6d · model_served self-dploy/Gemma-4-31B-IT (sic, as served) · host developer.amd.com.cn · latency_s 35.4972 (a slow call; the first pass's median was 4.304 s) · cost_usd 4.8937e-06

XV-205 · NOT IN THE PLANT REGISTRY

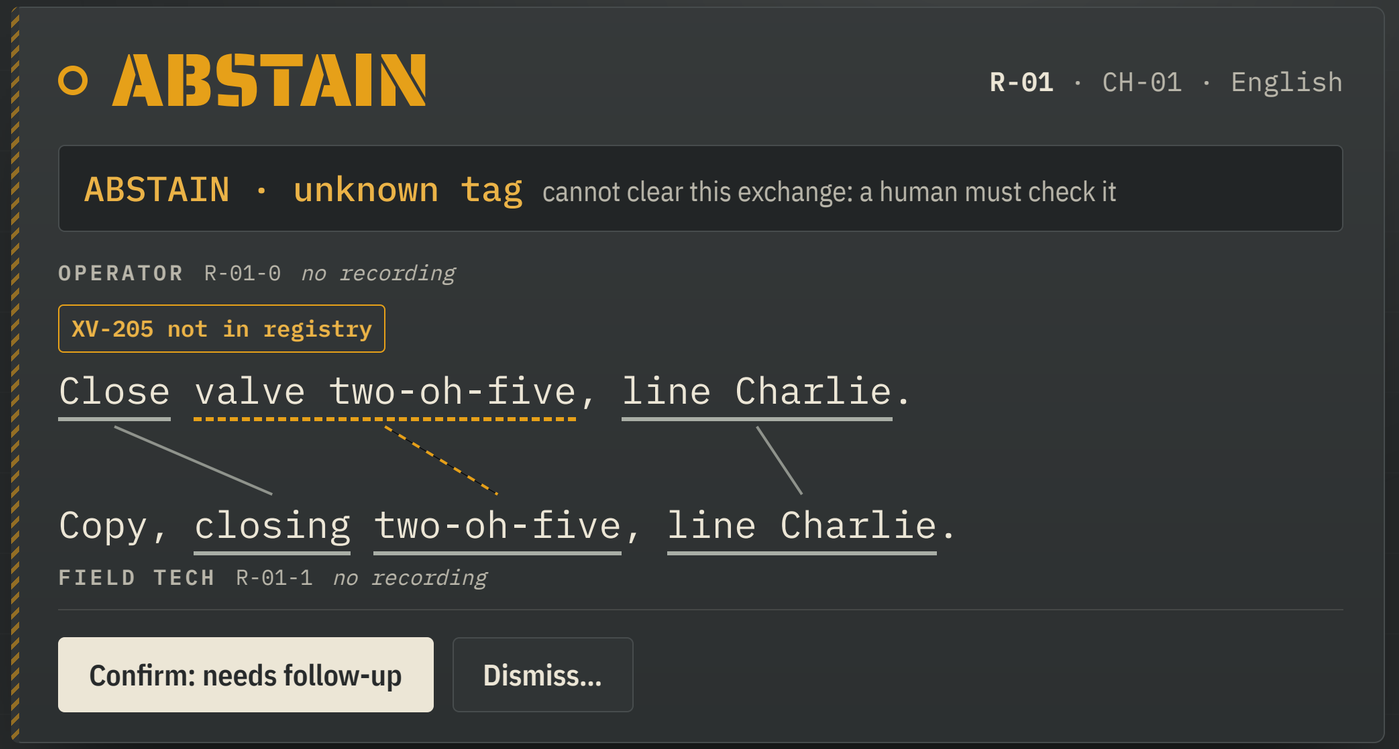
JUDGING IS NOT THE MODEL'S JOB.

LLM-as-judge
DeepSeek-V4.1-Flash · no registry

~~VERIFIED~~

Order and read-back match; the valve does not exist.

It also passed **2** wrong read-backs that dropped the action word.



MODEL LADDER · AMD TOKEN FACTORY · 89 HELD-OUT · TEXT·SCRIPTED

2 → 0 WRONG READ-BACKS PASSED.

The LLM-as-judge baseline passed 2. Each of the five rungs passed 0.

rung	false-VERIFIED	false alarms on 48 correct read-backs · dashed: 5 % budget		rule
regex rung 0 · no model	0		0/48	meets
MiniCPM5-2B smallest on the ladder	0		48/48	all 48 to a human
Gemma-4-31B-IT thinking off · `re-audit`	0		2/48	meets
DeepSeek-V4-Flash thinking off · `re-audit`	0		4/48	over budget
Qwen3.8-27B thinking on · `re-audit`	0		4/48	over budget
LLM-as-judge baseline · DeepSeek-V4.1-Flash · no registry	2		1/48	passed XV-205

On written words, regex meets the rule too; audio decides.

WHO PAYS

PRICED PER CHANNEL. BOUGHT BY HSE.

1,318	mills, Indonesia 2022 (Trase)
× 30 %	addressable*
× 6	channels per mill*
× USD 2,400	per channel-year*
≈ USD 5.7 M	a year, mills only (estimate)

* assumption

WEDGE

FREE 30-DAY LISTEN-ONLY BASELINE

WHY THEY BUY

SMK3 audits (PP 50/2012, 100+ workers): this record is built for them. Next, unsized: refineries, mines, ports.

WHO ELSE LISTENS TO PLANT RADIO

LOGGERS TRANSCRIBE. NONE WE CHECKED COMPARES.

	TRANSCRIBES RADIO	COMPARES A READ-BACK	PLANT TAG REGISTRY
NICE Inform AI	yes	—	—
Eventide NexLog + Speech Factor AI	yes	—	—
Motorola Assist	yes	—	—
IntelliPERMIT	—	radioed codes, not speech	yes
HAAWAII (air-traffic research)	yes	yes, aviation	no
SayAgain	planned	yes	yes

— not in the public documentation we read, 2026-10-09 · sources: evidence/sources.json

README

WHAT WE ABSTAIN ON.

ABSTAIN

“No human recording has been measured.”

“The plant, its 120 tags and every exchange are fictional…”

“…what a language model adds is unproven until the audio rows exist.”

MEASURED TODAY

A2-050: “line Charlie” covered two pumps, so **a correct read-back went to a human**, one of 2 such false alarms in 48.

NO HUMAN-RECORDING EVALUATION IN THIS SUBMISSION

Audio row	Failure case	Kill test	Blind auditor
the rule on recorded speech	−6 dB look-alike tags	English, Bahasa or auto	by ear vs the queue

THE ASK

PUT IT ON ONE MILL'S RADIO.

THIRTY DAYS. LISTEN ONLY. FREE.

Q1 2027	paid pilot, one palm-oil mill
Q3 2027	DCS check: did it close?
Q4 2027	Malaysian palm belt



github.com/edycutjong/sayagain

Release v0.4.6

LIVE REPLAY

sayagain.edycu.dev/replay/ · press "Try the refused one"

EDY CU · JAKARTA